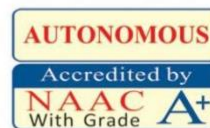




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Tutorial – ENGINEERING PHYSICS – MIC 23

25-26 -II Sem -II – MID

UNIT III

SHORT ANSWER QUESTIONS

1. Define Interplanar spacing and write its expression.
2. Problem on Interplanar spacing
3. State and explain Bragg's law and hence deduce the equation.
4. Bragg's Law problem.

LONG ANSWER QUESTIONS

1. Derive an expression for inter planar spacing between parallel planes in a cubic crystal.
2. Explain Bragg's X-Ray diffractometer.
3. Explain the Laue method of X-ray diffraction to determine crystal structure.
4. Explain powder method for studying crystal structure.

UNIT IV

SHORT ANSWER QUESTIONS

1. Explain deBroglie's hypothesis and derive the expressions for deBroglie's wavelength.
2. What are matter waves? Write the properties of matter waves.
3. State Heisenberg's uncertainty principle.
4. Explain the physical significance of wave function and write the conditions for a physically acceptable wave function (limitations of wave function)
5. Discuss the merits and drawbacks of Classical free electron theory.
6. What are the salient features or assumptions of Quantum free electron theory?
7. Problems on de Broglie's wave length.

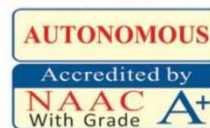
LONG ANSWER QUESTIONS

1. Derive Schrodinger's time independent and time dependent wave equations .
2. Discuss the behavior of an electron moving in a one dimensional potential well and obtain an expression for its energy.
(or) Apply Schrodinger wave equation to obtain the expression for energy of a particle in a one dimensional potential well and show that energy is quantized. (or) Analyse the behavior of a particle in a one dimensional potential well using Schrodinger's time independent wave equation and show that energy of the particle confined to that well is quantized.
3. Problems on energy of a particle.
4. Explain Fermi Dirac distribution and its temperature dependence. Illustrate graphically.



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UNIT V

SHORT ANSWER QUESTIONS

1. Distinguish between conductors, semiconductors and insulators based on energy gap.
2. Write the differences between intrinsic and extrinsic semiconductors.
3. Give examples of trivalent and pentavalent atoms.
4. Derive Einstein's relations.
5. Mention the applications of Hall effect.
6. Problem on Hall effect.

LONG ANSWER QUESTIONS

1. Derive an expression for carrier concentration of intrinsic semiconductor(n_i). (10M)
2. Explain the variation of fermi energy level with temperature in intrinsic and extrinsic semiconductor.(or)
Analyse the dependence of Fermi level on concentration of charge carriers in an extrinsic semiconductor.(or)
Discuss how Fermi energy in n-type and p-type semiconductors is affected by impurity concentration and temperature with figures.
3. What are drift and diffusion currents and derive their expressions.
4. State and explain Hall effect and derive an expression for Hall coefficient.